MAZEN EMAD ABDELHALIM

Agentic AI Engineer | LLM & RAG Systems | Multi-Agent Orchestration

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PROFESSIONAL SUMMARY

AI Engineer with end-to-end experience across the full AI lifecycle, from data engineering and classical ML to modern LLMs and agentic AI. Proven ability to build production-ready multi-agent platforms that orchestrate tool use and manage retrieval-augmented memory over vector databases. Deeply experienced across the full RAG stack and NLP systems design. Highly skilled in taking projects from raw data to deployed solutions, grounded in robust software engineering practices including REST API design, automated testing, and containerized deployment. A fast learner adept at translating emerging AI research into scalable, real-world automation solutions.

EDUCATION

B.Sc. in Artificial Intelligence | Benha University – Faculty of Computers and Artificial Intelligence, Benha, Egypt
2022 – 2026 | GPA: 3.54 / 4.0 | Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Data Mining, Algorithm Design

TECHNICAL SKILLS

- Agentic AI & Orchestration: LangGraph, CrewAI, LangChain, multi-agent workflow design, tool/API integration, custom tool building, proactive/notification-based human-in-the-loop workflows
- RAG & LLM Engineering: Retrieval-Augmented Generation, vector databases, prompt engineering, hybrid search (BM25 + dense), CrossEncoder reranking, embeddings, model evaluation, LLM API integration (OpenAI, Google Gemini, Groq, Cohere, Ollama/LLaMA)
- Data Science, ML & Deep Learning: Data preprocessing & feature engineering, supervised/unsupervised ML, deep learning (CNN, RNN/LSTM, Transformers), NLP (tokenization, text classification, BERT fine-tuning), transfer learning, model evaluation & optimization
- Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, LangChain/LangGraph, Pandas, NumPy, Pydantic (schema validation)
- Software Engineering: Python (production-level), FastAPI & REST API design, Git/GitHub version control, Pytest automated testing, OOP, Data Structures & Algorithms
- Deployment & MLOps: Docker & Docker Compose, async task processing (Celery + Redis), MongoDB, model deployment & monitoring, observability logging
- Other: SQL, Agile methodologies, technical documentation, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Junior Data Scientist / AI Engineer | X Clan *Jul 2025 – Oct 2025*

- Developed and fine-tuned machine learning models for predictive analytics on production-scale data (500,000+ records, 100+ engineered features), boosting baseline accuracy by 15% to an overall performance of 92%.
- Collaborated on containerizing and deploying models into production using FastAPI and Docker, delivering sub-second latency for real-time inference requests at scale — direct experience with the API design, deployment, and reliability practices needed to ship AI solutions into production.
- Automated data preparation and preprocessing pipelines in Python, reducing preprocessing time by 30% while maintaining production-level code quality and Git version-control discipline.

RELEVANT PROJECTS — AGENTIC AI, RAG & END-TO-END AI SYSTEMS

NexusAI — Autonomous Multi-Agent AI Secretary
2026 | github.com/MazenEmad14/NexusAI

- Architected a multi-agent assistant orchestrated with LangGraph on a FastAPI backend, defining agent routing, tool integration, and decision logic to autonomously manage email, calendar, and knowledge-base queries.
- Integrated multiple external tools and APIs (Gmail, Google Calendar, web search, voice STT/TTS) into the agent's toolset, and built a RAG knowledge base over Qdrant for grounded, semantic-search-based generation.
- Implemented a proactive background scheduler that evaluates incoming emails and triggers human-in-the-loop notifications, plus a Pytest test suite and Docker-based MongoDB setup for reliable, repeatable deployment.

WhatsApp RAG Agent — Production Retrieval-Augmented Generation System
2026 | github.com/MazenEmad14/WhatsApp_Agent

- Built a production-style RAG assistant delivering grounded answers natively in WhatsApp, combining hybrid retrieval (BM25 + dense vector search via Qdrant) with CrossEncoder reranking for high-precision context selection.
- Deployed a fully containerized, self-hosted pipeline (Docker for Qdrant/Redis, async Celery workers) integrating the Groq LLM API and local GPU-accelerated embedding/reranking models — demonstrating end-to-end automation-platform and deployment engineering.
- Added an observability layer to log agent interactions and implemented zero-downtime knowledge-base reindexing via vector-store aliasing, reflecting responsible, production-grade system design.

Hematology AI Diagnostic Agent — Multi-Agent Orchestration (Graduation Project)
2025 – 2026 | github.com/MazenEmad14/Grad_Agent

- Orchestrated 3 specialized deep learning models (vision, OCR, structured data) through a LangGraph agent state machine, defining task routing and decision logic to autonomously drive a clinical diagnostic workflow end-to-end.
- Implemented one of the pipeline's core agents as a multi-modal fusion architecture in PyTorch — an EfficientNet-B0 CNN backbone (transfer learning) fused with a tabular head for structured lab data — to automate blood-disorder detection, reducing manual analysis time by an estimated 70%.
- Deployed a RESTful API (FastAPI) exposing the multi-agent pipeline for diagnostic inference and automated report generation, achieving 100% uptime under production API constraints — demonstrating the ability to take a multi-agent system from design to a deployed, end-to-end product.

CERTIFICATIONS & TRAINING

- Generative AI Engineer — DEPI (2026): Multi-agent systems, RAG pipelines, and production AI deployment/monitoring using open-source and hosted LLMs.
- Generative AI Code Camp — ITI & NVIDIA (2026): 35-hour intensive program on generative AI fundamentals, applied LLM application building, and RAG-based augmentation.
- Natural Language Processing — NTI (2025): Built NLP pipelines and fine-tuned Transformer models (BERT) on 10,000+ documents.
- Machine Learning for Data Analysis — NTI (2025): EDA and feature engineering on 20,000+ record datasets; optimized supervised/unsupervised models to 90%+ accuracy.
- Artificial Intelligence Fundamentals — ITI (2025): Designed and trained DNNs and CNNs, achieving 85%+ validation accuracy on benchmark datasets.

LANGUAGES

Arabic: Native Proficiency | English: Professional Working Proficiency (B2)